ENEX12002

LECTURE NOTES

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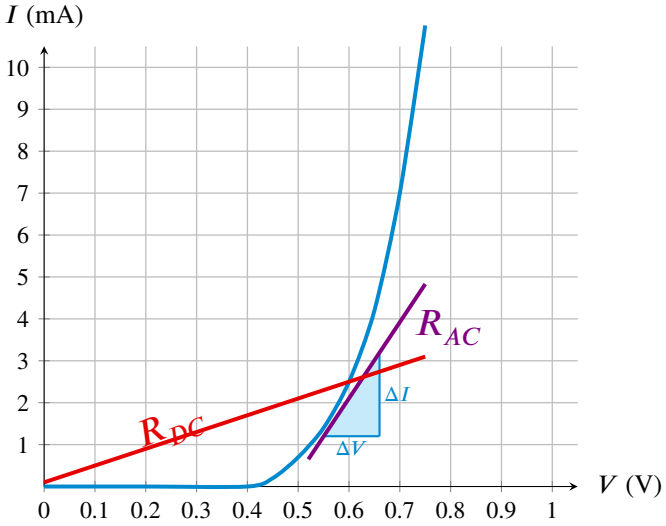
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1 Analog Circuits

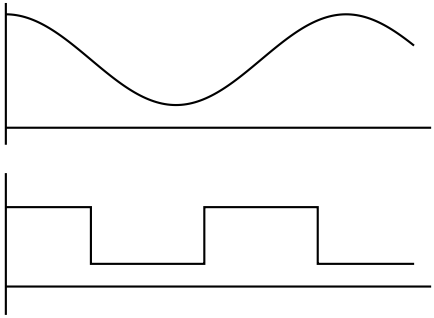
1.1 Characteristic Curves:

A characteristic curve is a type of linear plot that demonstates a relationship between two variable properties of a device. Most electronic devices CC refers too a plot of current as a function of coltage. Mostly known as an I-V curve.



1.2 Analog vs Digital Signals:

Signalling within a circuit is generally classified as either continuous or discrete. Continuous (Analog signalling) varies after a range of values within limits. Discrete (Digital signals) change between two discreet voltage levels.

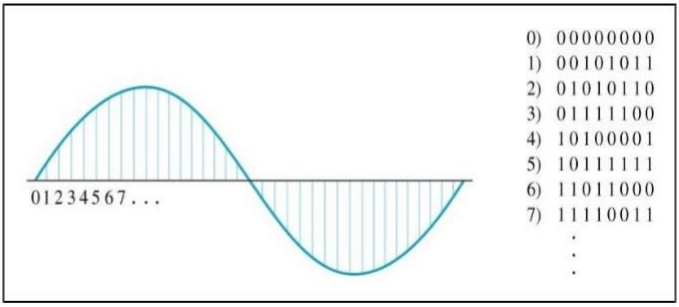


1.3 Analog & Digital Circuits:

Analog circuits are used in situations implementing linear functions, wave shaping and transformation between current and voltage. Digital circuit however ase used to process data rapidly and in calculations. Most systems utilize a mix of both systems (e.g AV systems).

1.4 Sampling:

When converting between signal types sampling must take place. Sampling is the approximation of analog's original waveform into time slices. The disadvantage of sampling is that is loses some precision, however the advantages of a digital system out weight this. The physical process of sampling is called quantizing. This produces numbers that can be processed by digital systems.



1.5 Vectors:

A vector is a line that contains both direction and magnitude. An AC circuits sinusodial waveform can be generated by plotting the project of a rotating vector in circular motion.

1.6 Frequency and Period:

The frequency of a periodic wave is the number of cycles that occur every second. The SI unit for this is Hz. The period is how long it takes a waveform to complete one cycle.

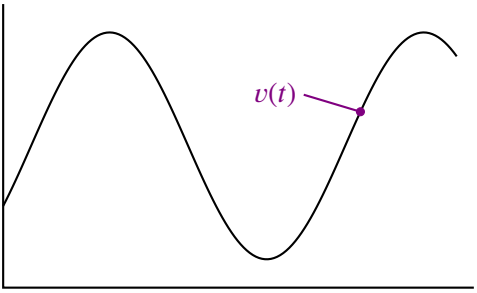
$$f = \frac{1}{T}$$

$$T = \frac{1}{f}$$

1.7 Instantaneous Values:

$$v(t) = V_p \sin(\omega t \pm \phi)$$

The instantaneous values are always against time (t).



1.8 Waveform Values:

The amplitude is the maximum distance of the values from the line of equilibrium. The average value of a sine wave is the found in $|v(t)|$, then averaging the curve.

$$V_{ave} = \frac{2V_p}{\pi} \approx 0.637V_p$$

1.9 RMS Values:

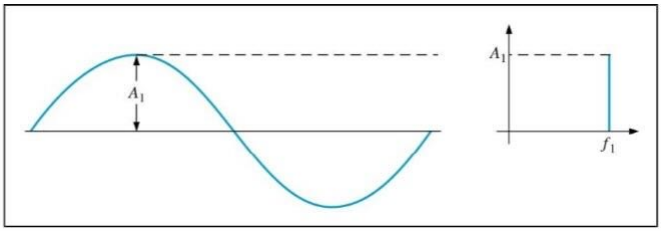
In order to compare AC and DC currents and voltages, AC is converted into its DC equivalent DC heating value. RMS current and voltage are their peak values.

$$\begin{aligned} V_{RMS} &= \frac{V_p}{\sqrt{2}} \\ &\approx 0.707V_p \end{aligned}$$

$$\begin{aligned} I_{RMS} &= \frac{I_p}{\sqrt{2}} \\ &\approx 0.707I_p \end{aligned}$$

1.10 Domain Signals:

Time domain is the funcation against time. An example is a osccilisope which shows values over a period of time. The frequency domain - A sine-wave against time can be represented against its frequency. This representations shows the waveforms amplitude and frequency but not it's phase angle.



1.11 Transducers:

These are a device that converts a physical quantity (e.g pressure, position or temp) from one form to another. Their outupts are quite small and require amplification. Passice transducers require external power to functions, this is called excitation. Active transducers are self-generating that convert a portion being measured into electrical signal.

1.12 Amplification:

Amps are designde to amplify either voltage or power. Linear amplifiers produce a magnified replica of the inputted signal, producing a useful output.

$$A_v = \frac{V_{out}}{V_{in}}$$

1.13 Transfer Curves:

All amplifiers, when ideal, produce a linear transfer curves. Beyond this limit they no longer act idealy, this is called saturation.

1.14 Power Gain:

$$A_p = \frac{P_{out}}{P_{in}} = \frac{V_{out}I_{out}}{V_{in}I_{in}}$$

1.15 dB Power Ratios:

The decibal of a power ratio is:

$$\begin{aligned} A_p &= 10\log_{10}(\frac{P_1}{P_2}) \\ &= 10\log_{10}(A_p) \end{aligned}$$

1.16 dB Voltage Ratio:

$$\begin{aligned} A_v &= 20\log_{10}(\frac{V_1}{V_2}) \\ &= 20\log_{10}(A_v) \end{aligned}$$

2 Basic Circuit Theory

2.1 Electrical Charge:

Electrical effects are attributed to the seperation of charges and their movements. Electrical forces are that opposites attract and similar repels. A charge is considered continuous if $q = 1.6022E\text{-}19e^{-}$.

2.2 Voltage and Current:

Rather than working purely in charge, it is easier to work with it in voltage and current.

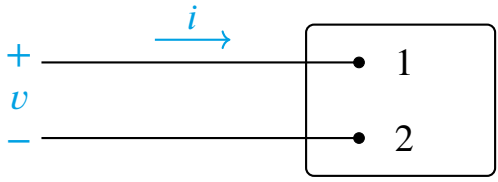
$$V = \frac{dw}{dq}$$

$$I = \frac{dq}{dt}$$

2.3 Ideal Basic Circuits:

These are made up of 5 main points:

1. Two Terminals
2. Can be described mathmatically in terms of voltage and current
3. Cannot be subdivided into further elements
4. Utilizes the passive sign convention

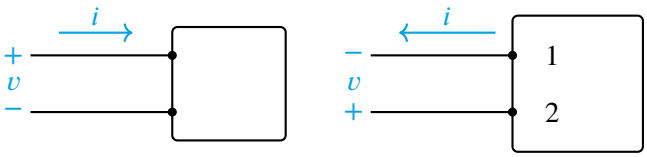


2.4 Power and Energy:

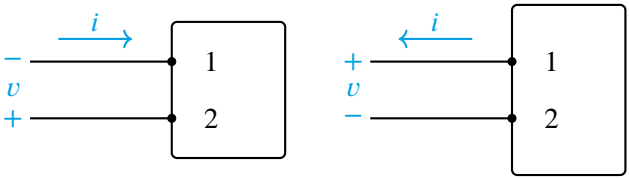
The useful output of a system is often non-electrical. Practical applications also have limitations in the amount of power it can handle.

$$p = \frac{dw}{dt} = (\frac{dw}{dq})(\frac{dq}{dt}) = VI$$

The polarity describes where current flows in the relationship to voltage. If $p > 0$, power is being delivered to the circuit inside the box. For instance a motor.



Else if $p < 0$, power is being extracted from a circuit. For example a battery.



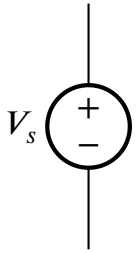
Elements that make up a basic circui include:

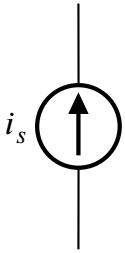
- Voltage source/s
- Current source/s
- Passive Circuit elements, As they do not produce electricity
  - Resistors
  - Inductors
  - Capacitors

2.5 Voltage and Current Sources:

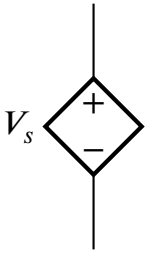
An electrical source that can convert between electrical and non-electrical energy. An ideal voltage source provides a consistant voltage regardless of current. This is similar for current sources. In real-world applications, ideal power sources do not exist.

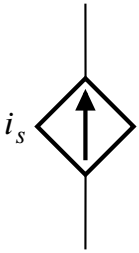
2.5.1 Source Types





Independent Sources





Dependent Sources

2.6 Kirchoffs Laws:

A circuit is said to be solved once a voltage and current is found across every circuit element.

$$\text{KCL} \Rightarrow \sum I_{out} = \sum I_{in}$$

$$\text{KVL} \Rightarrow \sum V_{loop} = 0$$

To apply these laws, polarity must be applied throughout the circuit.

2.7 Resistance:

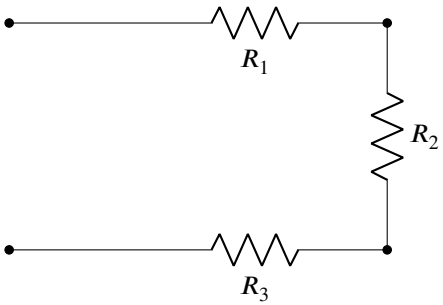
Resistors resist the flow of current within a circuit. Conductance is also the inverse of resistance, $G = \frac{1}{R}$.



$$V = IR$$

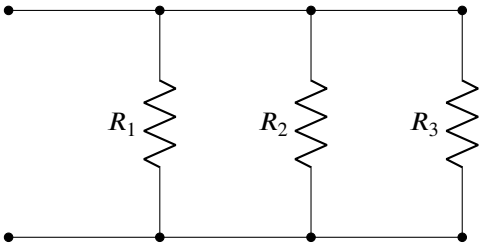
2.7.1 Series

$$R_{eq} = \sum_{n=1}^{k} R_n$$



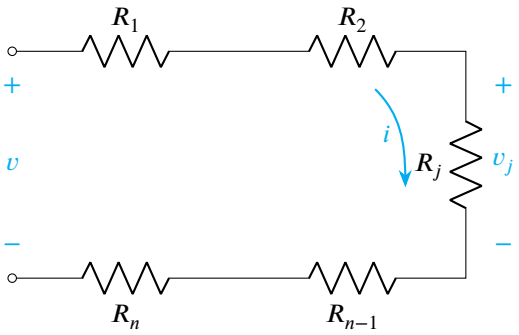
2.7.2 Parllel

$$R_{eq} = \frac{1}{\sum^{k} n = 1 \frac{1}{R_n}}$$



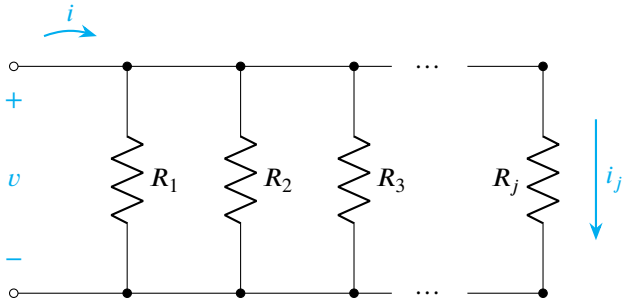
2.8 Voltage Division:

$$V_j = iR_j = \frac{R_j}{R_{eq}} V$$



2.9 Current Division:

$$i_j = \frac{V}{R_j} = \frac{R_{eq}}{R_j} i$$



2.10 Source Transformations:

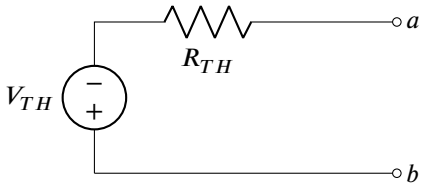
Current and voltage sources can be interchanged. This is because equivalent sources produce the same electrical effects, regardless of load.

2.10.1 Thévenin Equivalent

A way to simplify a circuit that focuses on the behaviour at the terminals.

$$V_{TH} = V_{O/C} = V_{ab}$$

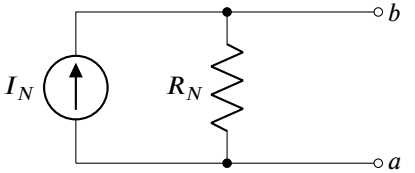
$$R_{TH} = \frac{V_{TH}}{I_{SC}}$$



2.10.2 Norton Equivalent

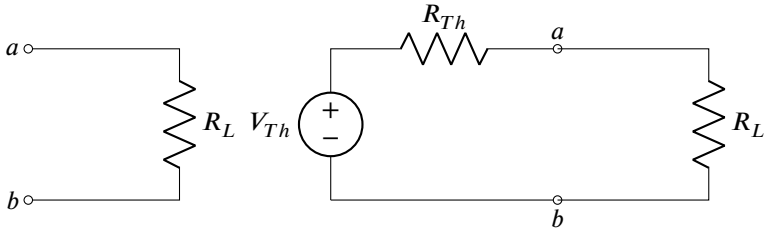
$$I_N = I_{S/C}$$

$$R_N = R_{TH}$$



2.11 Maximum Power:

To find a circuits maximum power, replace its resistive network within its Thévenin equivalent.



Power delivery to R_L:

$$p = I^2 R_L = (\frac{V_{TH}}{R_{TH} + R_L})^2 R_L$$

Max power is when $R_L = R_{TH}$:

$$p_{max} = \frac{V_{TH}^2 R_L}{(2R_L)^2} = \frac{V_{TH}^2}{4R_L}$$

2.12 Superposition Principle:

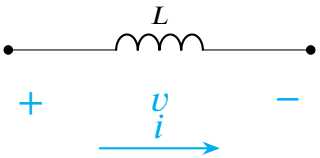
A linear system obeys this principle. Which is when two or more independent energy sources are connected in a circuit, the total response is the sum of each individual.

2.13 Inductors:

The behaviour of inductors are based around the properties of electro--magnetism. The SI unit for an inductor is Henry.

$$V = L\frac{di}{dt}$$

$$W = \frac{1}{2}LI^2$$



$$i(t) = \frac{1}{L}\int_{t_0}^{t} v\, d\tau + i(t_0)$$

$$p = Li\frac{di}{dt} = \frac{dw}{dt} = vi(t)$$

2.13.1 Series

$$L_{eq} = \sum_{n=1}^{k} L_n$$

2.13.2 Parallel

$$L_{eq} = \frac{1}{\sum_{n=1}^{k} \frac{1}{L_n}}$$

2.14 Capacitor:

Made-up of two conductors seperated by an insulator. It acts as an O/C to DC current and S/C to AC. An important note to remember that electrical charge is not directly transported through the capacitor.



$$V = IR$$

$$i = C\frac{dv}{dt}$$

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i\, d\tau + v(t_0)$$

$$p = vi = Cv\frac{dv}{dt}$$

$$w = \frac{1}{2}Cv^2$$

2.14.1 Series

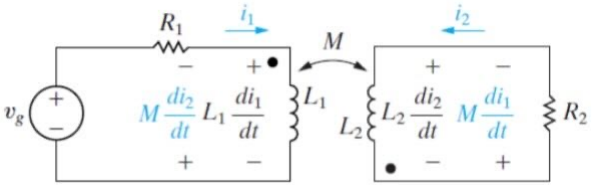
$$C_{eq} = \frac{1}{\sum_{n=1}^{k}\frac{1}{C_n}}$$

2.14.2 Parallel

$$C_{eq} = \sum_{n=1}^{k} C_n$$

2.15 Mutual Inductance:

The magnetic fields between the two circuits create a mutual and self-
-inductance voltages. Polarity of M depends on the way the coils are wound. L_1 and L_2 must be wound is opposing directions.



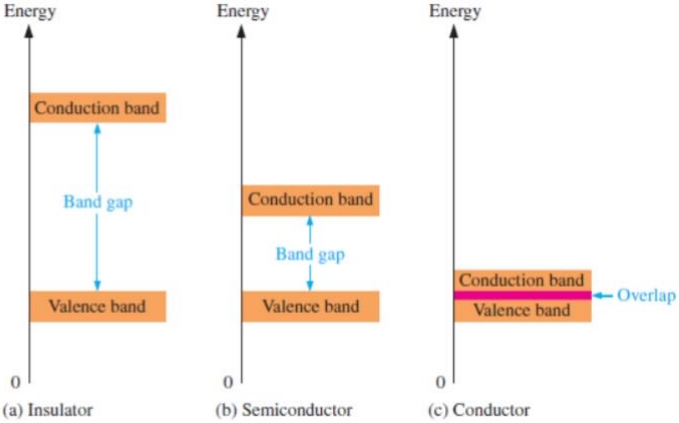
3 Diodes

3.1 Chemistry Revision:

Electrons form themselves into shells, with the outer shell containing the most energy due to their great mobility. These electrons in the outer shell are called valence electrons. Ions are formed through either an element donating or recieving electrons (e.g. Cations and Anions)

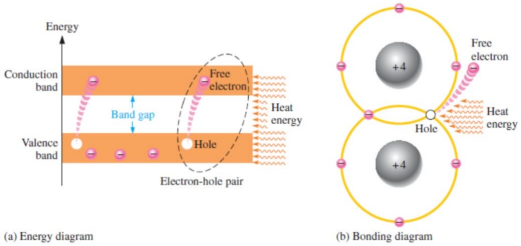
3.2 Energy Gaps:

Valence bonds within inter-elemental bonds are in the valence band. A conduction band is the energy band beyond the valence band.



Material	$\rho(\Omega \cdot cm)$
Insulator	$\rho > 10^5$
Semiconductor	$10^{-5} < \rho < 10^5$
Conductor	$\rho < 10^{-5}$

3.3 Electrons and Holes:



In instrinsic silicon it's current is produced by the movement of free electrons toward the end. The effective movement of the hole is opposite direction of electron movement.

3.4 Silicon:

3.4.1 Doping Silicon

3.4.1.1 Donor Atoms

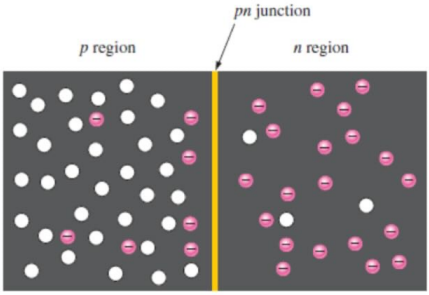
These atoms are pentavalent imparity, with 5 electrons in the valence shell. They are doped into the silicon crystal structure that donates the extra electron. This then becomes a conduction electron.

3.4.1.2 Acceptor Atoms

Place a trivalent imparity into the silicon crystal structure. Creating a hole within the silicon molecule.

3.4.2 PN Junction

In the image below, the acceptor is the *p*-region and donor is *n*-region. Within the PN-junction, the *p*-region is negatively charged, and the *n*-region is slightly positively charge.



The PN-junction forms a basic diode. This is something that only allows current flow in one direction. A voltage drop is formed across the reigon as depletion of *n*-type silicon and equillibrium occurs when it repel electron migration.

3.5 Diodes:

3.5.1 Forward Bias

An external voltage assist electrons in moving through the PN-junction, allowing higher current flow. This reduces the width of the PN-junction.

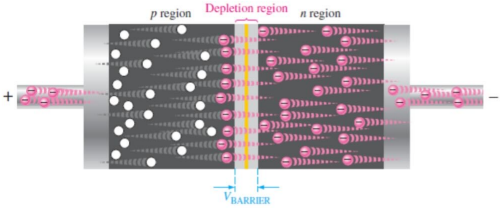
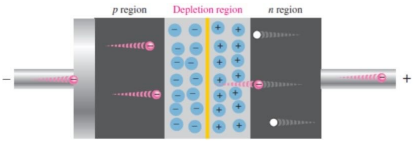


FIGURE 2–4
A forward-biased diode showing the flow of majority carriers and the voltage due to the barrier potential across the depletion region.

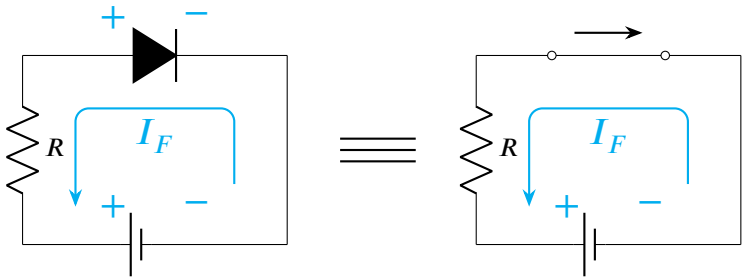
3.5.2 Reverse Bias

An extremely small reverse current pushes electrons back towards the PN--junction.



3.6 Ideal Diode Models:

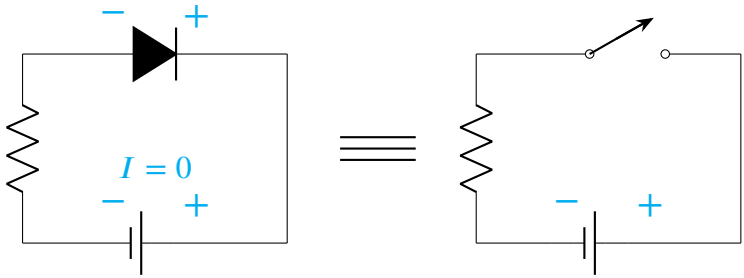
3.6.1 Forward Bias



$$I_F = \frac{V_{BIAS}}{R_{LIMIT}}$$

$V_F = 0V$, acts as S/C

3.6.2 Reverse Bias

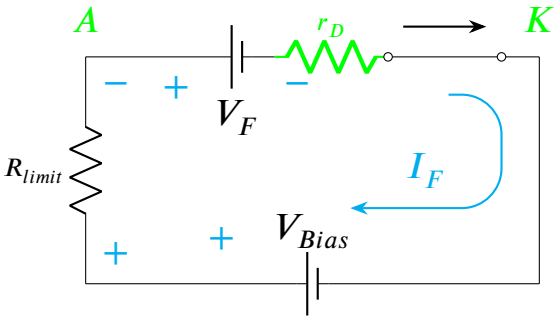


$$I_R = 0$$

$V_R = V_{BIAS}$, acts as O/C

3.7 Practical Diode Models:

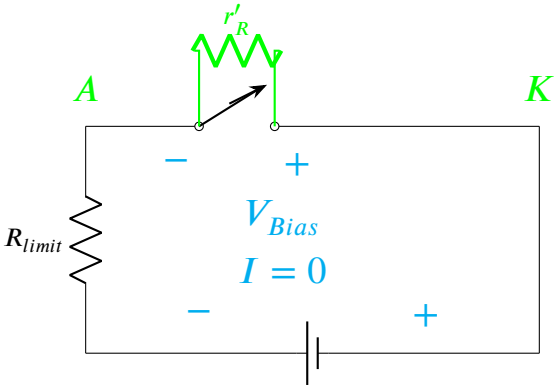
3.7.1 Forward Bias



$$V_F = 0.7 + I_F r'_d$$

$$I_F = \frac{V_{BIAS} - 0.7}{R_{LIMIT} + r'_d}$$

3.7.2 Reverse Bias

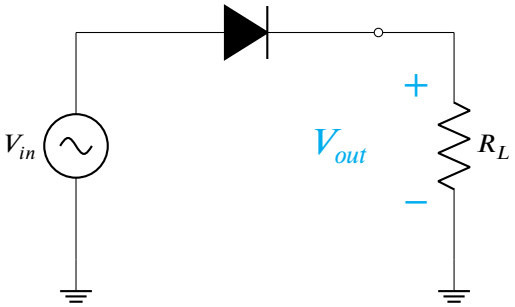


The cathode side of the diode is connected to positive end of the power supply. Whereas, the anode is connected to the negative side.

4 Diode Applications

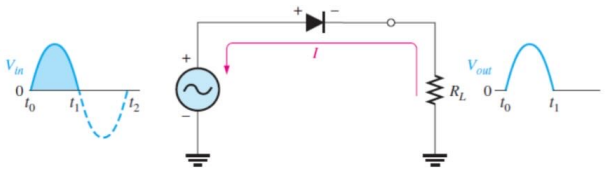
4.1 Wave Rectifiers:

A rectifieris a type of circuit configurations that converts AC to its pulsating DC equivalent. A half-wave rectifier only converts half of the AC sine wave (e.g. positive or negative). A full wave rectifier converts the entirety of the sine wave.

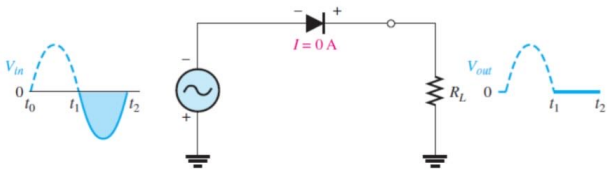


For the circuit above, if V_{in} is negative or smaller than the voltage barries in the diode, the diode should be reversed bias.

4.1.1 Half-wave operation



(a) During the positive alternation of the 60 Hz input voltage, the output voltage looks like the positive half of the input voltage. The current path is through ground back to the source.



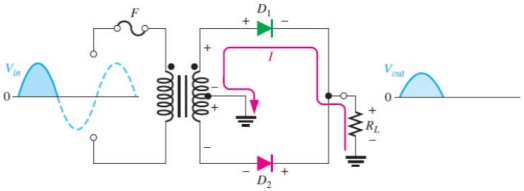
(b) During the negative alternation of the input voltage, the current is 0, so the output voltage is also 0.



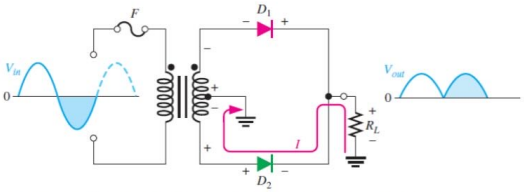
(c) 60 Hz half-wave output voltage for three input cycles

4.1.2 Full-wave Rectifier

This circuit controls the direction of the load currents so that only the positive or negative outputs are recieved.

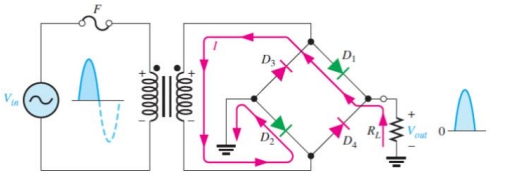


(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.

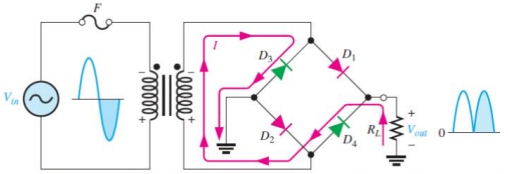


(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

4.1.3 Bridge Rectifier



(a) During the positive half-cycle of the input, D_1 and D_2 are forward-biased and conduct current. D_3 and D_4 are reverse-biased.

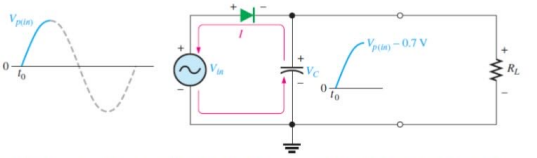


(b) During the negative half-cycle of the input, D_3 and D_4 are forward-biased and conduct current. D_1 and D_2 are reverse-biased.

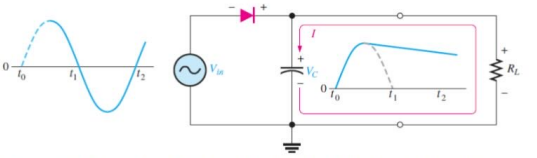
4.1.4 Smoothing Rectified Signal

The ripple within the output signal can put strain on connected DC components. A capcitor is used to smooth the signal when the voltage no longer charges the

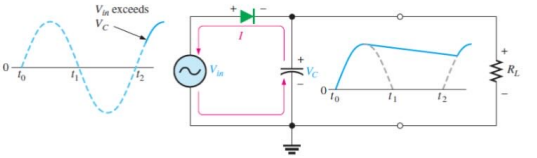
capacitor. The larger the capacitor the small the ripple, and smoother the output voltage will be.



(a) Initial charging of the capacitor (diode is forward-biased) happens only once when power is turned on.



(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.



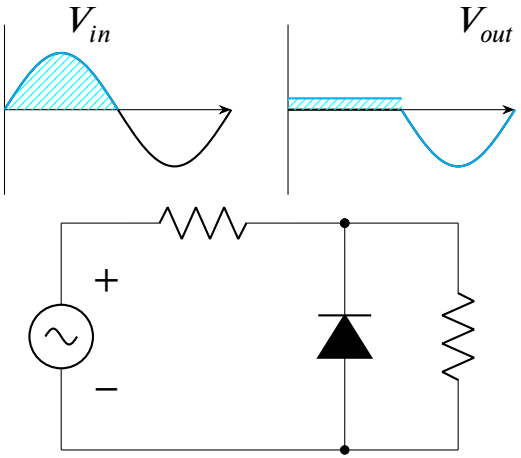
(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid dark blue curve.

4.2 IC Regulators:

An intergrated circuit (IC) is a complete circuit on a small chip of silicon. The IC regulator is an IC connected to the output of a rectifier maintaining a consistent output voltage. The most common ones are 3-pin. The pins are for input, output, and reference adjustment.

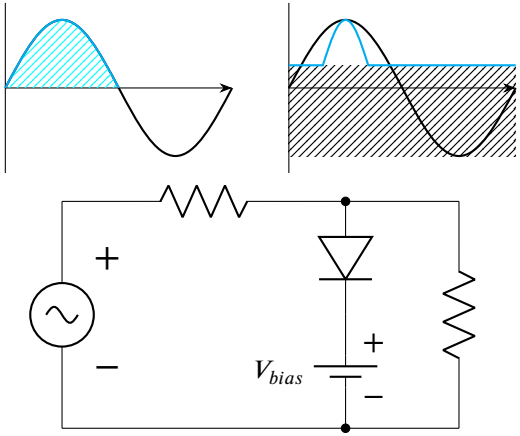
4.3 Diode Limiters:

Limits the voltage to a certain value within the diode itself. The voltage is only limited to the circuit parallel to the diode. A common example of this is the Zener diode.

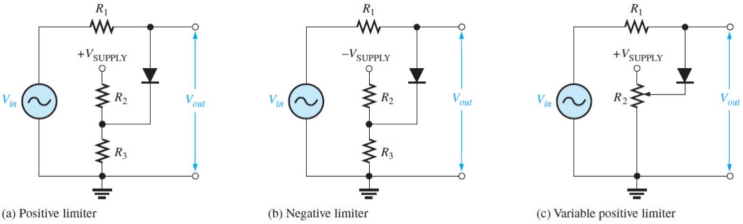


4.4 Bias Limiters:

Similar to a diode limiter, but maintains a stable minimum voltage.

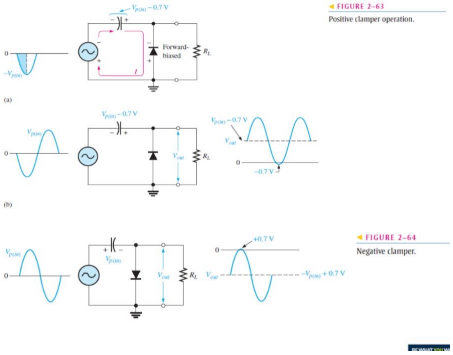


4.5 Bias Variation:



4.6 Diode Clampers:

Adds a DC offset to an AC waveform. These can also be known as DC restorers.



4.7 Extras:

PIV of a diode is just the complete negative of the input voltage. As the diode acts like a S/C. All diodes in series drop 0.7V per diode.
V_{RMS} of half wave rectifier is:

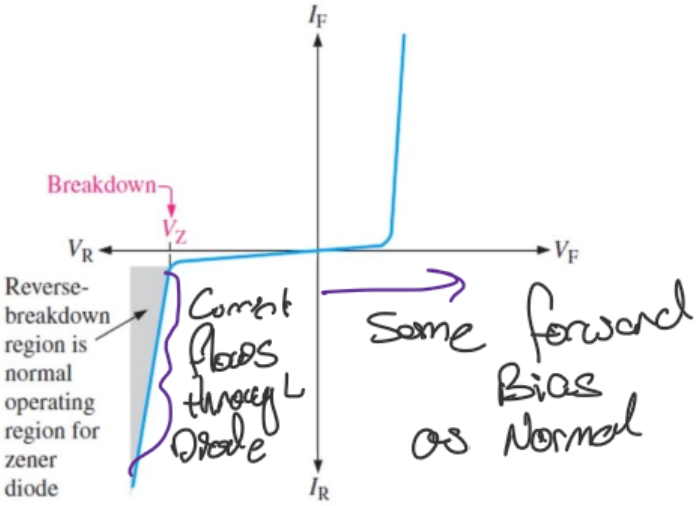
$$V_{RMS} = \frac{1}{\sqrt{2}} \cdot \frac{V_{out}}{2}$$

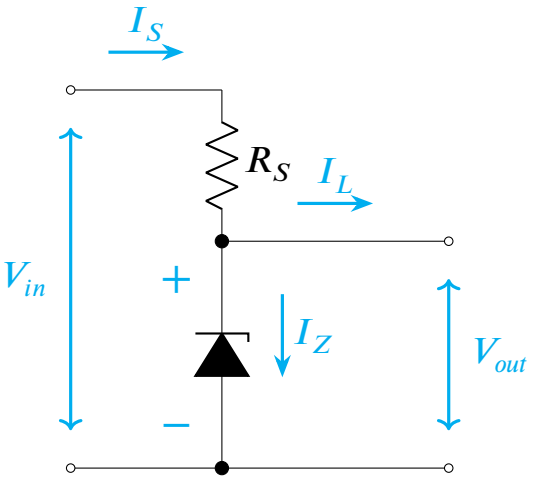
$$V_{avg} = \frac{4V_{out}}{\pi}$$

5 Special Diodes

5.1 Zener Diode:

A Zener diode is one which mainly operates in its reverse break-down region.





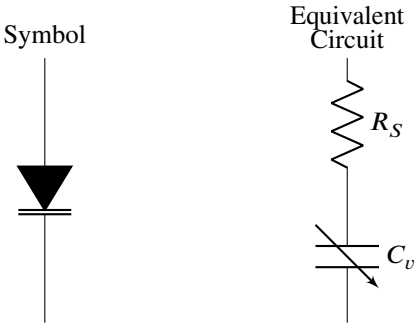
$$R_S = \frac{V_{in} - V_{out}}{I_Z}$$

$$I_{Z(load)} = I_{Z(noload)} - I_L$$

The breakdown voltage, is the ideal values for voltage items. After the maximum current through the diode. If more current is applied to the diode, permenant damage occurs. The knee current is the minimum current required for the voltage regulation to work properly.

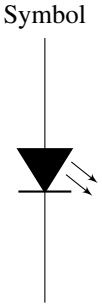
5.2 Varactor Diode:

This diode is a reversed-biased PN-junction that utilizeds the inherent capacitence of the depletion region.



5.3 LightEmtting Diodes:

LEDsemit light when exposed to power. As such they are very useful as indicator throughoug a circuit. These diode types are designed to work only in forward bias conditions.



5.3.1 Organic LED:

The way OLED works is identical to the normal LEDS. The make-up of an OLED display is 2-3 layers, composed of organic polymers that emit light on

certain voltage levels.

5.4 Photodiode:

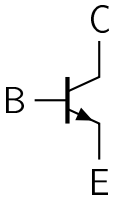
A ligh absorbing device that conducts current based on light levels. This works in reverse bias configuration.

Symbol

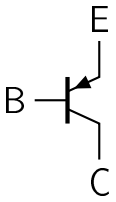


6 Bipolar Junction Transistor

The following symbols are used to denote between the two types present. In where the types of silicon are placed within the silicon.



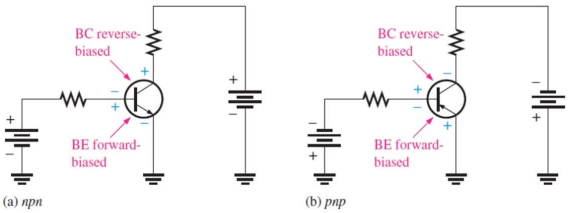
npn



pnp

6.1 BJT Biasing:

For a BJT to operate properly, it must be biased. The collector must be in reverse bias and emitter forward. The current direction opposes the direction the arrow is pointing.



$$I_E = I_B + I_C$$

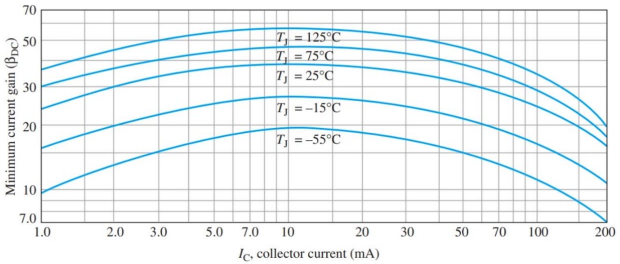
Since base current is much lower than emittor or collector current the can equation equates to roughly $I_E \approx I_C$.

6.2 Beta DC:

β_{DC} is the current gain of a transistor. On a datashett β_{DC} is usually h_{FE}. This is only valid if the transistor operates within the manufacters specified range.

$$\beta_{DC} = \frac{I_C}{I_B}$$

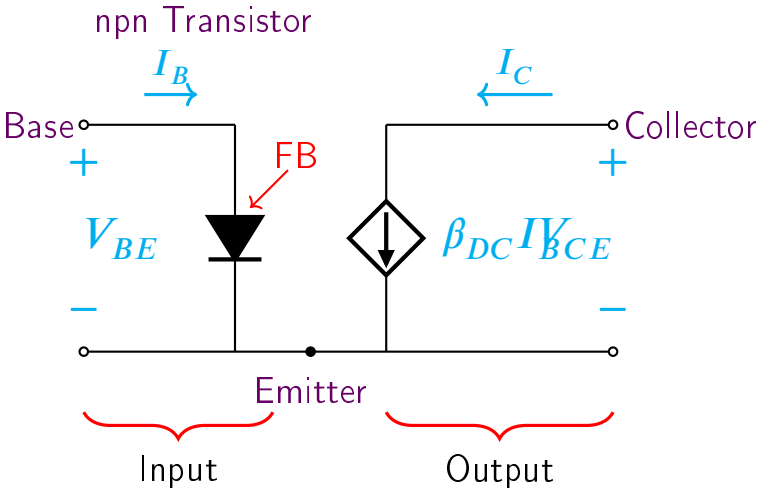
β_{DC} is not always a consistent value, as shown in the image below.



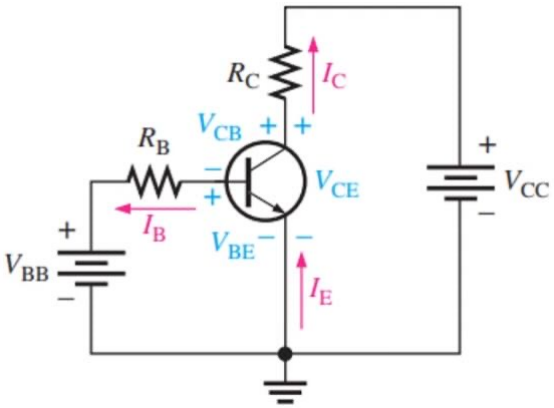
6.3 Alpha DC:

α_{DC} is the ratio of the DC collector to DC emittor currents. Typical value ranges for α_{DC} is $0.95 \rightarrow 0.99$. It is less common for α_{DC} to be found in the components datasheets than β_{DC}.

6.4 Equivalent Circuits:



6.5 Voltage and Current:



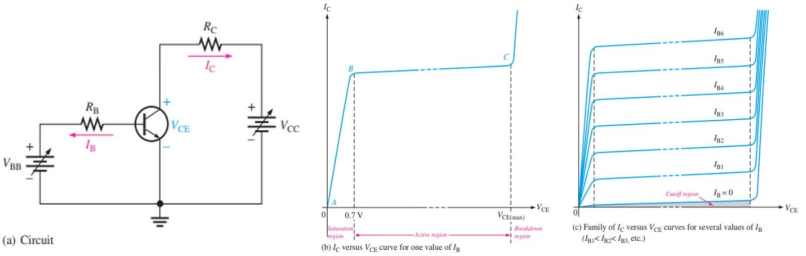
$$V_{BE} = 0.7V$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CB} = V_{CE} - V_{BE}$$

6.6 Collector Characteristic Curves:



6.7 Cut-off:

When $I_B \approx 0$, the BJT is cut off and there is no collector current. In cut-off both base-emitter and base-collector are in reverse bias.

6.8 Saturation:

When base current is large, the V_{CC} drop across R_C is sufficient, leaving no voltage between the collector and emitter.

$$I_{C(sat)} = \frac{V_{CC}}{R_C}$$

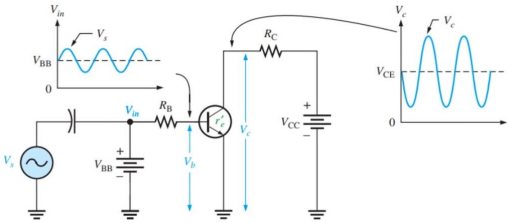
$$I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C}$$

6.9 Maximum Rating:

Transistors have limitations specified on the datasheet. As such the V_{CE} and I_C of the component must not exceed the $P_{D(max)}$. At higher temps the $P_{D(max)}$ is less.

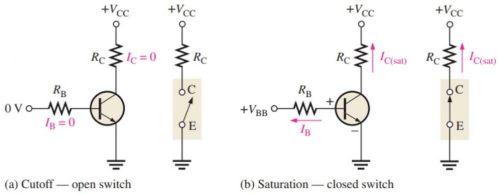
6.10 BJT as an Amplifier:

$$V_{out} = \frac{R_C}{r'_e} \times V_b$$



6.11 BJT as a Switch:

By switching a BJT in either cut-off or saturation, a basic switch can be formed. This is either through letting current not flow or flow through the transistor.

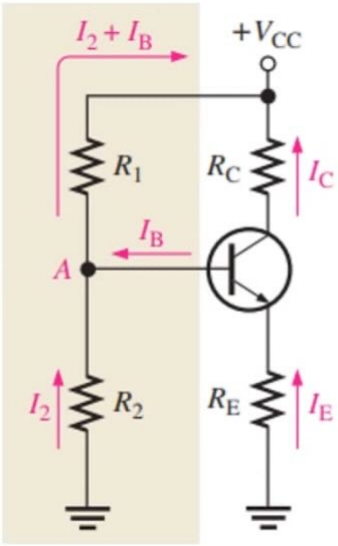


7 Transistor Bias Circuits

7.1 Q Point - DC Operating Point:

The way a BJT amplifies a signal depends on it's biasing of itself.

7.2 Voltage Divider Biasing:



$$
\begin{aligned}
V_B &\approx \frac{R_2}{R_1 + R_2} \times V_{CC} \\
V_{BE} &= V_B - V_E \\
&\approx 0.7V \\
I_C &\approx I_E \\
&= \frac{V_E}{R_E} \\
V_C &= V_{CC} - I_C R_C \\
V_{CE} &= V_C - V_E
\end{aligned}
$$

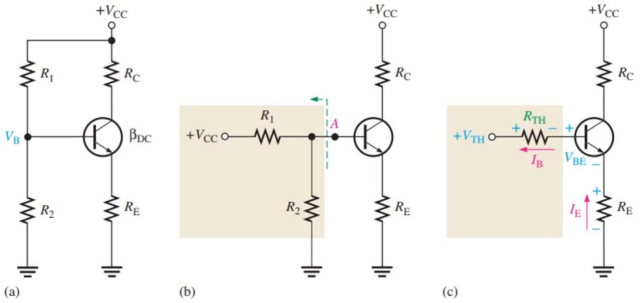
7.3 DC Input Resistance:

Important as it is the resistance presented to an input signal coming into the base.

$$
\begin{aligned}
R_{in(Base)} &= \frac{V_B}{I_B} \\
&= \frac{\beta_{DC} V_B}{I_E}
\end{aligned}
$$

As it is the reliant upon β_{DC}, it will vary for different transistors at different temperatures.

7.4 Voltage Division with Thevénin Equivalent Circuits:



$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{CC}$$

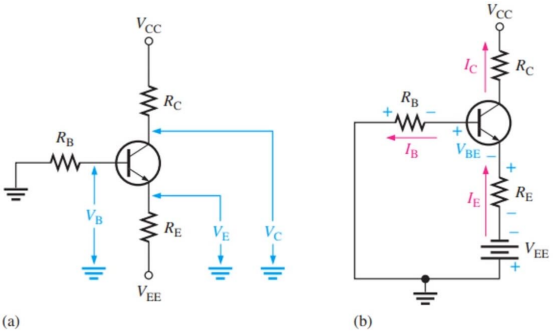
$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2}$$

$$V_{TH} - V_{R_{TH}} - V_{BE} - V_{R_E} = 0$$

$$V_{TH} = F_E(R_E + \frac{R_{TH}}{\beta_{DC}}) + V_{BE}$$

$$F_E = \frac{V_{TH} - V_{BE}}{R_E + \frac{R_{TH}}{\beta_{DC}}}$$

7.5 Emitter Biasing:



$$V_{EE} + V_{R_B} + V_{BE} + V_{R_E} = 0$$

$$-V_{EE} = \frac{I_E}{\beta_{DC}} R_B + I_E R_E + V_{BE}$$

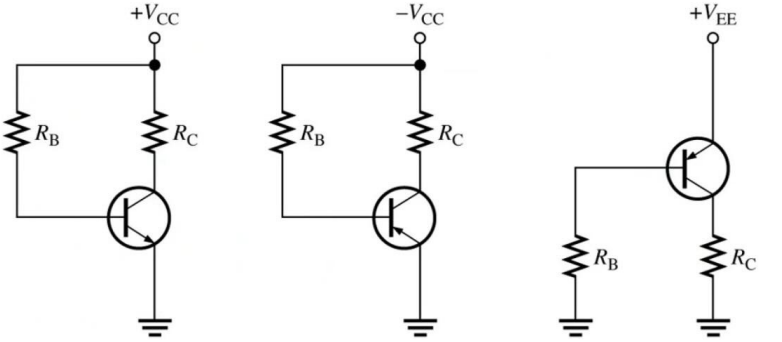
$$I_E = \frac{-V_{EE} - V_{BE}}{R_E + \frac{R_B}{\beta_{DC}}}$$

$$V_E = V_{EE} + I_E R_E$$

$$V_B = V_E + V_{BE}$$

$$V_C = V_{CC} - I_C R_C$$

7.6 Base Biasing:



(a) *npn* bias circuit

(b) *pnp* bias circuit with a negative supply

(c) *pnp* bias circuit with a positive supply

$$V_{CC} - V_{R_B} - V_{BE} = 0$$

$$V_{CC} - I_B R_B - V_{BE} = 0$$

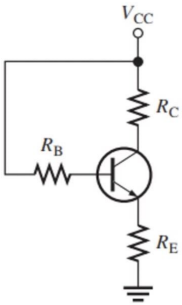
$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$I_C = \beta_{DC}\left(\frac{V_{CC} - V_{BE}}{R_B}\right)$$

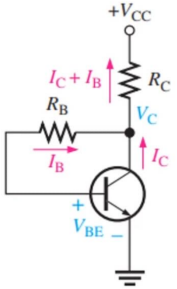
7.7 Emitter Feedback Biasing:



$$-V_{CC} + I_B R_B + V_{BE} + I_E R_E = 0$$

$$I_E = \frac{V_{CC} - V_{BE}}{R_E + \frac{R_B}{\beta_{DC}}}$$

7.8 Collector Feedback Biasing:



$$I_B = \frac{V_C - V_{BE}}{R_B}$$

$$= \frac{I_C}{\beta_{DC}}$$

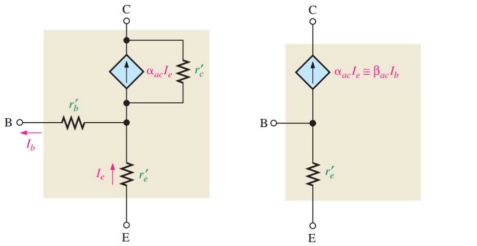
$$V_C \approx V_{CC} - I_C R_C$$

$$I_C = \frac{V_{CC} - V_{BE}}{R_C + \frac{R_B}{\beta_{DC}}}$$

$$V_{CE} = V_{CC} - I_C R_C$$

8 Bipolar Junction Transistors as Amplifiers

8.1 Tansistor AC Model:

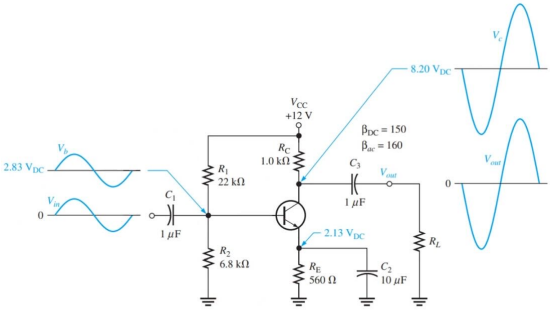


(a) Generalized *r*-parameter model for a BJT (b) Simplified *r*-parameter model for a BJT

Parameter	Description
α_{AC}	Alpha AC ($\frac{I_C}{I_E}$)
β_{AC}	Beta AC ($\frac{I_C}{I_0}$)
r'_e	AC Emitter Resistor
r'_b	AC Base Resistor
r'_c	AC Collector Resistor

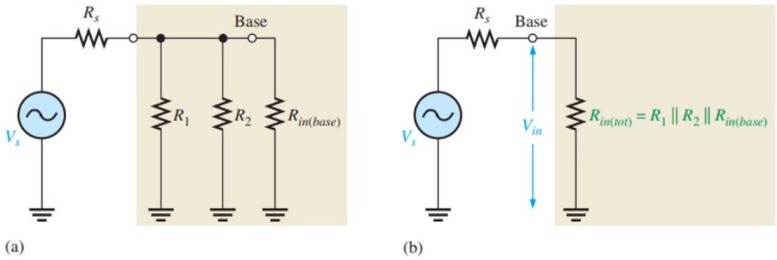
$$r'_e \approx \frac{25\text{mV}}{I_E}$$

8.2 Collector-Emitter Amplifier:



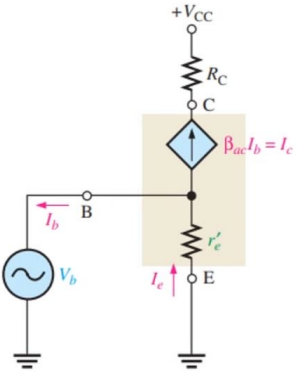
▲ FIGURE 6–8
A common-emitter amplifier.

Useful for very high voltage and current gains. To calculate the CE amplifier, complete a DC analysis to find Q-point and V_{CE}. After, do the AC analysis.



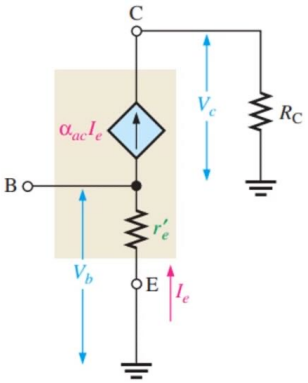
$$R_{in(tot)} = R_1||R_2||R_{in(base)}$$

$$V_b = (\frac{R_{in(tot)}}{R_s + R_{in(tot)}})V_s$$



$$\begin{aligned} R_{in(base)} &= \frac{V_{in}}{I_{in}} \\ &= \frac{V_b}{I_b} \\ &= \frac{I_e r'_e}{I_E \beta_{ac}} \\ &= \beta_{ac} r'_e \end{aligned}$$

$$\begin{aligned} R_{out} &= R_c||r'_e \\ &= R_C||R_L \end{aligned}$$

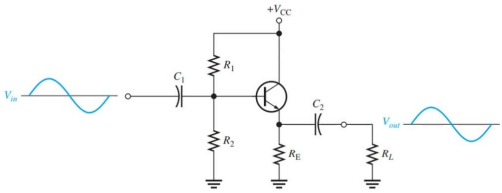


$$V_c = -\alpha_{ac} I_E R_c$$
$$\approx -I_E R_C$$
$$V_b = I_e r'_e$$
$$A_V = \frac{V_c}{V_b}$$
$$= -\frac{R_c}{r'_e}$$

The negative sign is used to indicate that the voltage is 180° out of phase.

$$A_i = \frac{I_c}{I_s}$$
$$I_s = \frac{V_s}{R_s + R_{in(tot)}}$$
$$A_P = A'_v A_i$$
$$A'_v = \frac{V_c}{V_s}$$

8.3 Emitter-follower Amplifier:



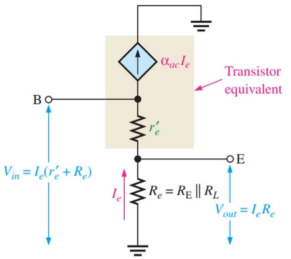
This type of amplifier is not great at voltage gain ($A_v \approx 1$), but allows high current gain.

$$V_{out} = I_e R_e$$

$$V_{in} = I_e(r'_e + R_e)$$

$$A_v = \frac{R_e}{r'_e + R_e}$$

When $V_{in} \approx V_{out}$, it has the same amplitude and phase-angle as the source.



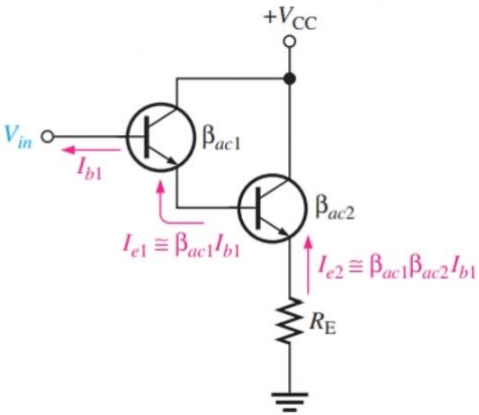
$$
\begin{aligned}
A_{in(base)} &= \frac{V_b}{I_b} \\
&= \frac{I_e(r'_e + R_e)}{I_b} \\
&= \beta_{ac}(r'_e + R_{eq})
\end{aligned}
$$

$$
\begin{aligned}
R_{in(tot)} &= R_1 || R_2 || R_{in(base)} \\
R_{out} &\approx \frac{R_s}{\beta_{ac}} || R_E
\end{aligned}
$$

$$
\begin{aligned}
A_i &= \frac{I_e}{I_{in}} \\
&= \frac{I_e R_{in(tot)}}{V_{in}} \\
A_p &= A_v A_i \\
&\approx A_i, \text{ as } A_v \approx 1
\end{aligned}
$$

8.4 The Darlington Pair:

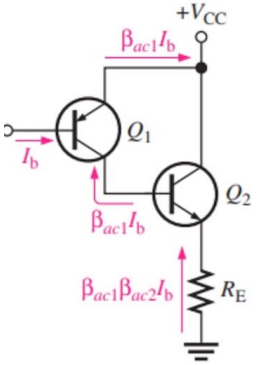
This is simply 2 cascading BJTs, with the emitter connected to the base of the other. The main advantage of using this is that it generates high input resistance and high current gain.



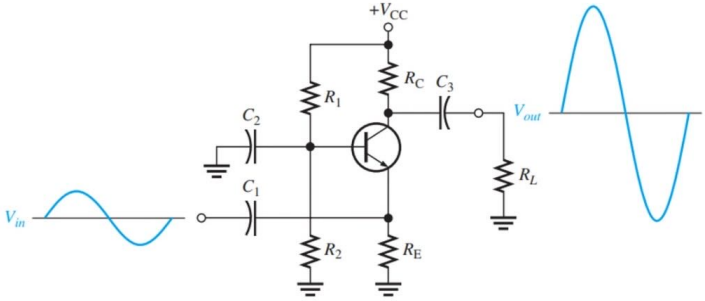
$$\beta_{ac} = \beta_{ac1}\beta_{ac2}$$

8.5 Szikali Pair:

Use both npn and pnp BJTs. Its main advantage is low turn on voltage. With the most common application being used in conjuction with the Darlington pair.

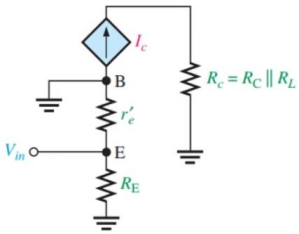


8.6 Collector-Base Amplifier:



(a) Complete circuit with load

$$\begin{aligned} A_v &= \frac{V_c}{V_e} \\ &= \frac{I_c R_c}{I_e(r'_e || R_e)} \\ &\approx \frac{R_c || R_L}{r'_e} \end{aligned}$$



(b) AC equivalent model

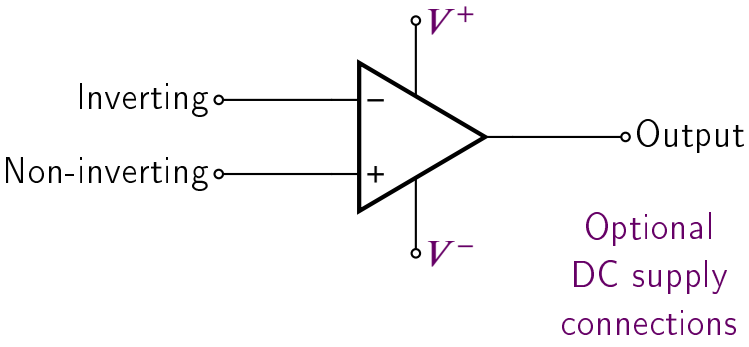
$$R_{in(emit)} = \frac{V_e}{I_e} = r'_e || R_e$$
$$R_{out} \approx R_e$$

$$A_v = \frac{r_{in}}{R_L}$$
$$A_i = 1$$
$$A_p \approx A_v$$

9 Operational Amplifiers

9.1 What is it?:

An OpAmp is name operational amplifier because they were initially used for mathematical operations. An OpAmp physically is a high DC amplifier, with extremely high voltage gain, high input impedance and low output impedance.



9.2 Ideal Op-Amp Characteristics:

1. Infinite Voltage Gain
2. Infinite Input Impedence
3. Zero Output Impedence

9.3 Practical Characteristic:

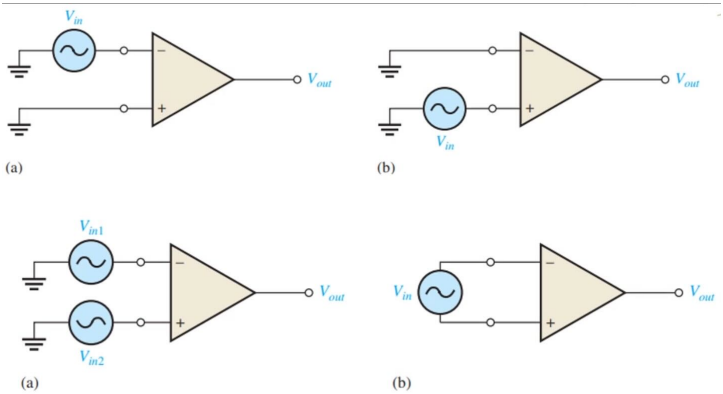
- P-P output voltage is limited to slightly less then the difference between two supply voltages.
- Output current is limited by internal restrictions.
- High amount of noise.

9.4 Op-Amp Design:

A typical op-amp is constructed of three parts:

- Differential Amp
- Voltage Amp
- Push-pull Amp

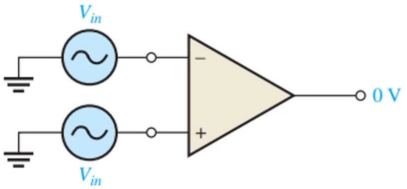
9.5 Differential Input Mode:



9.6 Open-Loop:

- Ratio of output to input voltage with no external components.
- Set by interal design.
- Usually a large value, 200,000 (106dB).
- Not a well controlled parameter.

9.7 Common-Input Mode:



$$CMMR = \frac{A_{ol}}{A_{cm}}$$

$$CMMR(dB) = 20\log_1 0(CMMR)$$

It is generally considered that a higher CMMR the better.

9.8 Maximum Output Voltage:

With no input signal the output should ideally be $0V$, and ideal limit of P-P of $\pm V_{CC}$. In practice $V_{out(P-P)} < \pm V_{CC}$ varying with the load connected and increasing directly with load resistance.

9.9 Input Offset Voltage:

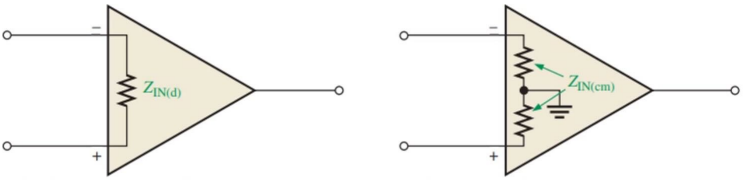
The ideal Op-Amp produces $0V_{out}$ for $0V_{in}$. However, practically a small DC voltage, $V_{out(error)}$ is created. On the datasheet, the input offset voltage (V_{os}), is the voltage required to force an output. Typically $V_{os} < 2mV$. A key note is that V_{os} drifts with temperature.

9.10 Input Bias Current:

The input terminals of a bipolar differential amplifier are transistor bases. The I_{BIAS} is the DC current required to properly operate the first stage.

$$I_{BIAS} = \frac{I_1 + I_2}{2}$$

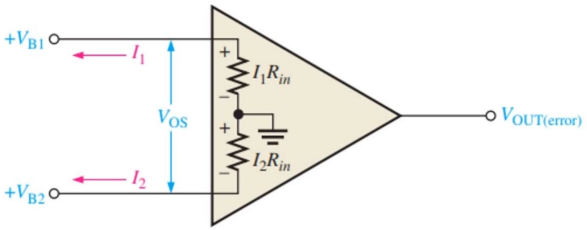
9.11 Input Impedence:



(a) Differential input impedance (b) Common-mode input impedance

Differential impedance is measured by determining the change in I_{BIAS}, for a gain change in V_{in}. THe total resistance between the inverting and non-inverting inputs. Common-mode is deremined through the change in bias for common-mode voltage input resistance between each input and ground.

9.12 Input Current:

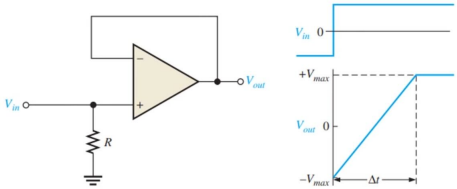


$$I_{OS} = |I_1 - I_2|$$
$$V_{OS} = I_{os} R_{in}$$
$$V_{out(error)} = A_v I_{os} R_{in}$$

9.13 Output Impedance:

Just the resistance viewed from the output terminal of the Op-Amp.

9.14 Slew Rate:



The max PoC of V_{out} is response the a step input voltage. The following equation gives the worst case senario.

$$\text{Slew Rate} = \frac{\Delta V_{out}}{\Delta t}$$

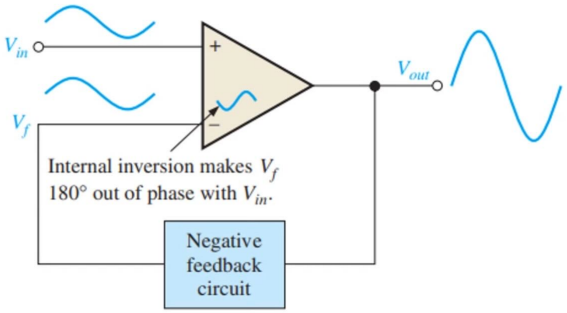
9.15 Noise Specification:

Noise in a circuit is an unwanted signal that affect a desired outcome. It is important for low-level signals and high accuracy operations. At low frequencies, noise ($\frac{1}{f}$) is often called pink noise. Above the critical noise frequency it becomes flat, called white noise.

$$\text{Noise} = \frac{V}{\sqrt{\text{Hz}}}$$

10 Operational Amplifier Feedback

10.1 Negative Feedback:



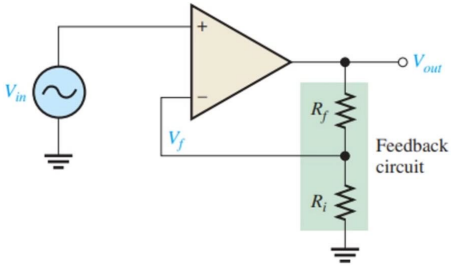
Useful in OpAmp circuits as they have very high internal voltage gain with small input signal. Driving the OpAmp toward saturation. Lowering is maximum amplification value.

10.2 Closed-loog Voltage Gain:

The A_{cl} si the A_v with external feedback, usually negative. The A_{cl} is determined by external component values and can be precisely controlled by them.

10.3 Non-Inverting Amplifier:

In this circuit the input and output signals are still in the same phase. It is made up of an OpAmp with an input signal applied to its non-inverting input. With some of the output place in the inverting side.

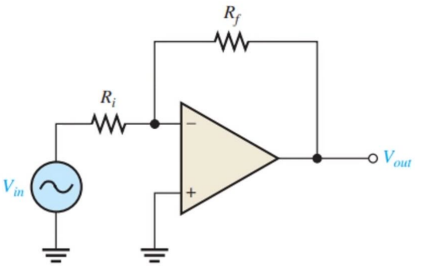


$$V_f = \frac{R_i}{R_i + R_f} \times V_{out}$$

$$B = \frac{R_i}{R_i + R_f}$$

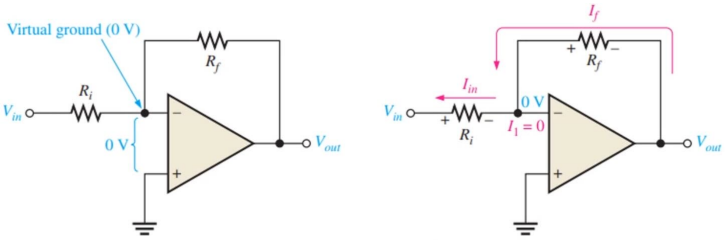
$$A_{cl(NI)} = \frac{V_{out}}{V_{in}} \approx \frac{1}{B}$$

10.4 Inverting Amplifier:



V_{in} will appear O/C as such current through the OpAmp is 0.

$$I_i = I_f$$



(a) Virtual ground

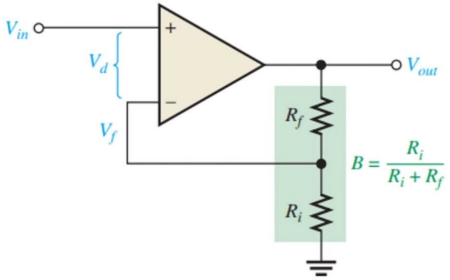
(b) $I_{in} = I_f$ and current at the inverting input (I_1) is 0.

$$I_{in} = \frac{V_{in}}{R_i}$$

$$I_f = \frac{-V_{out}}{R_L}$$

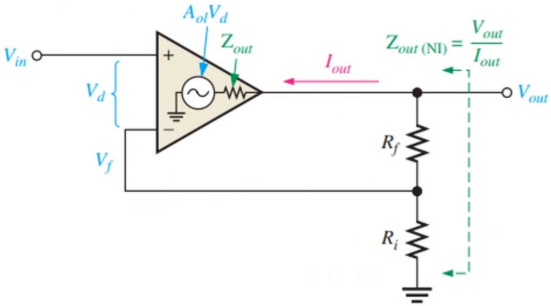
$$A_v = \frac{V_{out}}{V_{in}} = \frac{-R_f}{R_{in}}$$

10.5 Non-inverting Input Impedance:



$$\begin{aligned} V_{in} &= V_{ol} + V_f \\ &= V_d + A_{ol} B V_d \\ \frac{V_{in}}{I_{in}} &= (1 + A_{ol} B) Z_{in} \\ Z_{in(NI)} &= (1 + A_{ol} B) Z_{in} \end{aligned}$$

10.6 Non-inverting Output Impedance:



$$\begin{aligned} V_{out} &\approx A_{ol}(V_{in} - V_f) \\ &\approx A_{ol}(V_{in} - BV_{out}) \\ Z_{out} &= (1 + A_{ol}B)Z_{out(NI)} \\ Z_{out(NI)} &= \frac{Z_{out}}{1 + A_{ol}B} \end{aligned}$$

10.7 Voltage Follower Impedance:

Using $B = 1$:

$$Z_{in(VF)} = (1 + A_{ol})Z_{in}$$

$$Z_{out(VF)} = (\frac{Z_{out}}{1 + A_{ol}}$$

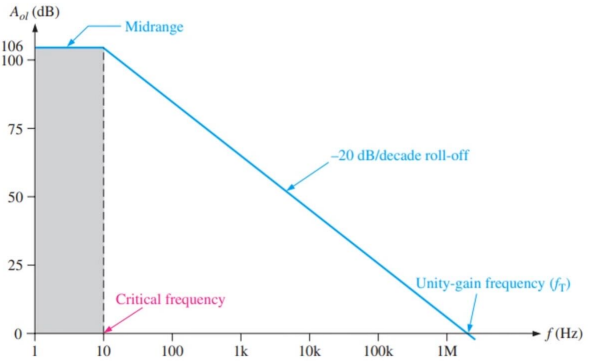
As such the input impedance of a voltage follower is greater than a non-inverting OpAmp voltage divider feedback network. Output is same as above but smaller than the other.

10.8 Inverting Impedances:

$$Z_{in(I)} \approx R_i$$

$$Z_{out(I)} = \frac{Z_{out}}{1 + A_{ol}B}$$

10.9 Bandwidth Limitations:

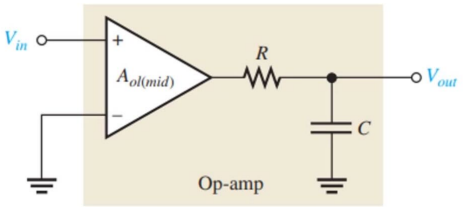


The bandwidth of the OpAmp is the frequency range between critical frequencies, where gaiin is 3dB lower then mid-range gain.

$$BW = f_{cu} - f_{cl}$$

10.10 Open-loop Gain vs Frequency:

THe RC lag (low-pass) circuits with an OpAmp are responsible for the roll-off in gain.



$$\frac{V_{out}}{V_{in}} = \frac{X_C}{\sqrt{R^2 + X_C^2}}$$
$$= \frac{1}{\sqrt{1 + \frac{f^2}{f_c^2}}}$$

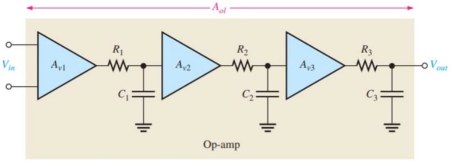
$$f_c = \frac{1}{2\pi RC}$$
$$A_{ol} = \frac{A_{ol(mid)}}{\sqrt{1 + \frac{f^2}{f_c^2}}}$$

The RC circuit causes a propagation delay from input to ouput. This causes some phase-shift between two signals.

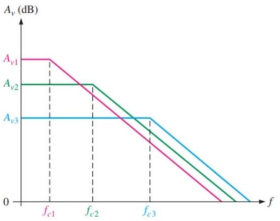
$$\theta = -\arctan\left(\frac{R}{X_C}\right)$$
$$= -\arctan\left(\frac{f}{f_C}\right)$$

10.11 Overall Response:

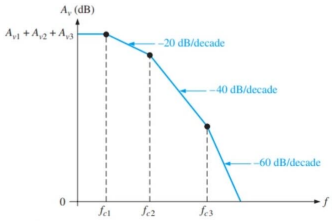
When multiple amplifiers are cascaded in stages, the total response is a composite of individual responses. The roll-off state are additive.



(a) Representation of an op-amp with three internal stages

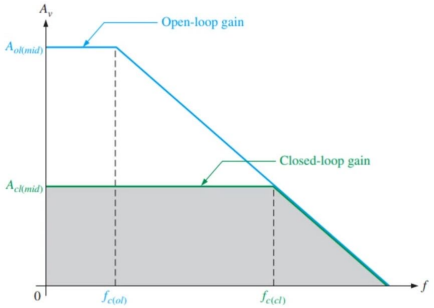


(b) Individual responses



(c) Composite response

10.12 Close-loop Gain:



$$
\begin{aligned}
A_{cl(NI)} &= \frac{A_{ol}}{1 + A_{ol}B} \\
&\approx \frac{1}{B} \\
&= 1 + \frac{R_f}{R_i} \\
A_{cl(I)} &\approx -\frac{R_f}{R_i} \\
f_{c(cl)} &= f_{c(ol)}(1 + BA_{ol(mid)}) \\
BW_{cl} &= BW_{ol}(1 + BA_{ol(mid)}) \\
A_{cl}f_{c(cl)} &= A_{ol}f_{c(ol)}
\end{aligned}
$$